Section III: Mechanisms of Action (The Logic Gates)

III. The Mechanism of Action

Definition 10: The Privacy Logic Gate (The Red/Green Switch)

To solve the computation objection (Panopticon), the system operates on a localized variance trigger. Privacy is the default state of low entropy.

The Transmission Function:

$$T(N_i) = \begin{cases} \text{NULL (Private)} & \text{if } \Delta < \epsilon \land \text{Var}(V_s) > \text{noise\_floor} \\ \text{FULL\_DUMP (Public)} & \text{if } \Delta \geq \epsilon \lor \text{Var}(V_s) \leq \text{noise\_floor} \end{cases} \tag{1}$$

Where:

- Δ: Divergence between Semantic Identity (V_i) and Somatic Reality (V_s).
- $\text{Var}(V_s) \leq \text{noise\_floor}$: Detection of artificial suppression (The "Dead Sensor" or "Drug" check).

Definition 11: The Thermodynamic Veto (The Policing Equation)

The Fiduciary State is prohibited from intervening if the energetic cost of intervention exceeds the entropic cost of the event. This serves as the check against "Over-Policing."

Let S_{event} be the entropy generated by a deviation. Let E_{police} be the energy required to correct it.

$$Action_{state} = \begin{cases} \text{IGNORE} & \text{if } S_{event} < E_{police} + E_{friction} \\ \text{INTERVENE} & \text{if } S_{event} \geq E_{police} + E_{friction} \end{cases} \tag{2}$$

Result: Petty friction is tolerated as "Thermal Noise." Only structural damage triggers mobilization.

Definition 12: The Neighbor Gradient Protocol (Anti-Parasitic Check)

To identify low-entropy predators (Psychopaths/Cult Leaders) who maintain $S_{int} \approx 0$ by exporting entropy to others, the system calculates the divergence of the local field.

$$LEI = \nabla \cdot S_{neighborhood} = \sum_{j \in \text{proximity}} \frac{dS_j}{dt} \tag{3}$$

The Predator Condition:

$$\text{If } S_{int}(N_i) \to 0 \wedge LEI(N_i) > \text{Threshold} \implies \text{Quarantine } N_i \tag{4}$$

Implication: A node cannot claim "peace" if its neighbors are burning.

Definition 13: The Analog Audit (The Ground Truth)

To prevent sensor decoupling (hacking), the system employs a stochastic physical verification protocol driven by a Quantum Random Number Generator (QRNG).

$$P(\text{Audit}|t) = \text{QRNG}(t) \tag{5}$$

Since $P(\text{Audit})$ is non-deterministic, the expected cost of sustained deception (E_{lie}) approaches infinity over time.

$$\lim_{t \to \infty} E_{lie} = \infty \tag{6}$$